

linnest

a typst interface to the `linnet` crate

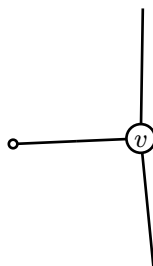
`linnet` is a rust crate that provides a graph data structure and many graph algorithms, and crucially for this package, multiple layout algorithms, as well as a dot parser (using the `dot_parser` crate).

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Linnest

The `linnet` crate that this package wraps around, is built around the half-edge graph data structure. This means that instead of a graph being represented as a set of nodes and edges, it is represented as a set of half-edges H , and a set of vertices V . The graph structure is then encoded through two maps. The first map, $\partial : H \rightarrow V$ maps each half-edge to its corresponding vertex. The preimage of any vertex v is the set of half-edges that map to it, called the crown of v . The second map, $\iota : H \rightarrow H$, is an involution that *glues* half edges together to form edges. If a half-edge is glued to itself, we call that an external half-edge. This means that `linnet` graphs are strictly more capable than normal edge and vertex graphs.



A side effect of natively supporting half-edges, is that subgraphs can be more granular, as we can encode them as sets of half-edges. This naturally supports vertex induced subgraphs (the union of the crowns of a set of vertices).

Linnest

Linnest is the `typst` plugin + API to integrate with `linnet` and access some of its capabilities. This means laying out algorithms, and dot parsing (all through `wasm`, so no dependencies/ external tooling required) along with a selection of graph algorithms (that can be expanded upon as needed).

Graphs can be constructed in two ways, parsing from a dot string using `parse`:

```
#let g = parse("digraph { a -> b }")
```

or building from edges and nodes using build, with a fletcher inspired syntax:

```
#let g = build({
  node(<a>, label: [$v$])
  node(<b>)
  edge(source(<a>), <a-b>, sink(<b>), label: [e])
  edge(source(<a>), <in-a1>, label: [e])
  edge(source(<a>), <in-a2>, label: [e])
})
```

In either case, type(g)=bytes, because they both return an opaque zero-copy value that corresponds to a linnet graph struct. However none of the data is lost, as you can query this value for information about the graph, such as the edges and the nodes.

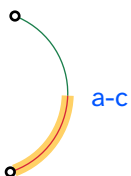
The main usecase however is of course to automatically place the nodes and edges on a canvas, using the layout function.

- graph for construction, parsing, inspection, joins, and graph algorithms.
- subgraph for subgraph object construction and inspection.
- layout for the separate layout pass.
- draw for rendering a laid-out graph object with CeTZ.
- physics for reusable particle-line edge styles.

Minimal Build Example

```
#import "../src/lib.typ": draw, graph, layout, subgraph
#import graph: build, dot, edge, edges, node, nodes, parse, sink, source
```

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>, compass: "e"),
    <a-c>,
    sink(<c>, compass: "w"),
    label: [a-c],
    statements: (
      color: "0055ff",
      source-color: "d72638",
      sink-color: "1b7f4c",
    ),
  ),
},
name: "demo",
)
#let g = layout(g)
#let east = subgraph.compass(g, "e")
#let edge-records = edges(g, subgraph: east)
#let dot-text = dot(g)
#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.label]
#let source-style(edge) = (stroke: rgb("#" + edge.source-color) + 0.5pt)
#let sink-style(edge) = (stroke: rgb("#" + edge.sink-color) + 0.5pt)
#draw(g, subgraph: east, edge-label: edge-label, source-style: source-style, sink-style: sink-style)
```



Graph Objects

Graph and subgraph objects are opaque zero-copy values. Build or parse graph objects with `graph`, transform graph objects with `layout`, and pass objects back to `graph` or `subgraph` for inspection.

- `graph.parse(input)` parses one or more DOT digraphs and returns an array of graph objects. Its `eval-graph-fields`, `eval-node-fields`, `eval-edge-fields`, `eval-source-fields`, and `eval-sink-fields` arguments are convenience arguments for `graph.eval-fields`.
- `graph.build(..)` constructs one graph object from a stream of node and edge items.
- `graph.map(graph, ..)` maps graph, node, edge, source, and sink records to new opaque payloads without changing topology.
- `graph.node-data(graph, <name>)` and `graph.edge-data(graph, <name>)` return one named node or edge payload.
- `graph.update-node-data(graph, <name>, data)` and `graph.update-edge-data(graph, <name>, data)` update one named node or edge payload. Direct replacements use a Rust-side lookup; callbacks still run in Typst with `(data, record)`.
- `graph.eval-fields(graph, ..)` evaluates selected record fields into payload entries. It works on parsed and built graph objects.
- `node(..)` returns a node item.
- `source(..)` and `sink(..)` return half-edge endpoints.
- `edge(..)` returns an edge item built from source/sink endpoints and an optional edge name.
- `layout(graph, ..)` runs layout as an explicit second step. Its settings are named parameters so calls stay descriptive and Tidy can document each field.
- `draw(graph, ..)` draws a laid-out graph object with CeTZ.
- `dot(graph)` returns a DOT string for inspection or export.

Graph Specs

The Typst construction API is half-edge first: create node items, create source and sink half-edge endpoints, then pass edge items to `graph.build`. `graph.build` accepts both comma-separated items and ordinary Typst code blocks. Typst labels such as `<a>`, `<h1>`, and `<e1>` are API names; they are resolved before the wire format is sent to the Rust plugin. Numeric `id` arguments are order/index overrides. The payload argument on graphs, nodes, edges, sources, and sinks is opaque Typst metadata: Typst CBOR-encodes it before the Rust plugin boundary, Rust archives the bytes without inspecting them, and Typst decodes it again in `graph.info`, `graph.nodes`, and `graph.edges`. The `label` convenience argument on nodes and edges is display content stored as `payload.label`; use statements: `(label: "...")` when a flat metadata label string is needed. Statements are flat metadata used by DOT; they cannot nest. Values are scalar strings/numbers/booleans. Use `payload` for structured Typst data or content.

When parsing DOT, the same payload channel can be filled from selected string fields. Evaluation happens in Typst at parse time; Rust only receives opaque payload bytes. The selected fields are evaluated with the record's merged fields dictionary in scope:

```
#let g = parse(  
  "digraph g { a [label=\"A\"]; a -> b [label=\"\${p}\", source=\"out\", sink=\"in\"] }",  
  eval-node-fields: ("label"),  
  eval-edge-fields: ("label"),  
  eval-source-fields: ("statement"),  
  eval-sink-fields: ("statement"),  
)  
#nodes(g).first().payload.label  
#edges(g).first().payload.label
```

The same transform can run after construction. This is useful for global edge statements that should apply to every edge while still seeing local edge fields:

```
#let g = build(  
  node(<a>)  
  node(<b>)  
  edge(source(<a>), sink(<b>), statements: (mom: "p"))  
, edge-statements: (display-label: "\${mom}"))
```

```
#let g = graph.eval-fields(g, eval-edge-fields: ("display-label",))
#edges(g).first().payload.at("display-label")

#let g = build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
```

Named nodes and edges can be updated after construction without scanning in the caller. The update replaces the opaque payload, or it can be a callback receiving (data, record):

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
#let g = update-node-data(g, <a>, (label: [A]))
#let g = update-edge-data(g, <e1>, (data, edge) => (
  label: [$p$],
  source: edge.source.node,
))
#node-data(g, <a>).label
#edge-data(g, <e1>).label
```

source(..) and sink(..) accept a node reference plus optional name, id, statement, and compass. name is a Typst label name for the half-edge; id is numeric:

```
edge(source(<a>, name: <h1>, id: 0), <e1>, sink(<c>, id: 2), label: [a-c])
```

One half-edge creates an external edge. The side is determined by the constructor, so there is no public flow argument:

```
edge(<incoming>, sink(<a>))
edge(source(<c>), <outgoing>)
```

String-valued statements may contain {name} placeholders. graph.build expands each placeholder while constructing the graph object. Use {{ and }} for literal braces. Unknown placeholders are left unchanged.

This is useful for graph-level edge-statements: the default edge statement is merged into each edge, then expanded against that edge's complete statement dictionary. The following default metadata records a string label derived from the per-edge label statement:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label: [a-c],
    statements: (color: "0055ff", label: "a-c"),
  )
},
edge-statements: (
  color: "000000",
  display-label: "{label}",
),
)
```

Placements

graph.pos creates a first-class placement. The default mode: "pin" turns a coordinate into a fixed layout constraint and a drawable position. Use mode: "start" when the coordinate should only seed the layout:

```
#let g = build({
  node(<a>, pos: graph.pos(x: -2, y: 0))
})
```

```

node(<c>, pos: graph.pos(ref: <a>, dx: 4, dy: 0))
edge(source(<a>), <a-c>, sink(<c>), pos: graph.pos(x: 0, y: 1.2))
})

```

graph.group links one coordinate across several nodes or edge control points. A side of "+" keeps the coordinate positive, and "-" keeps it negative. GammaLoop external-edge columns use this to keep incoming and outgoing external legs on opposite sides while pairing rows by a shared y group:

```

#let edge-items = (
  edge(
    source(<right-ext>),
    sink(<center>),
    pos: graph.pos(x: graph.group("right", side: "+"), y: graph.group("edgee0")),
  ),
  edge(
    source(<center>),
    sink(<left-ext>),
    pos: graph.pos(x: graph.group("left", side: "-"), y: graph.group("edgee0")),
  ),
)

```

Raw DOT input uses the same placement model through the pos attribute. The standard Graphviz subset is preserved: pos="x,y" is a starting coordinate and pos="x,y!" is pinned. Linnest extends the value with explicit id references and axis entries. In axis entries, ! belongs to that axis: numeric entries without ! are starting coordinates, numeric entries with ! are fixed constraints, and grouped entries must use ! because groups are constraints.

```

digraph {
  a [id=0 pos="0,0!"]
  b [id=1 pos="ref(node:0)+4,0!"]
  a -> b [id=0 pos="ref(node:1)+0,1!"]
  b -> c [id=1 pos="ref(edge:0)+1,0!"]
  c [id=2 pos="x:2!"]
  d [id=3 pos="x:2!,y:1"]
  ext -> a [id=2 pos="x:@-left!,y:@edge0!"]
}

```

The DOT grammar is intentionally id-based: ref(node:0) uses a node id and ref(edge:0) uses an edge id. Bare names and implicit edge order are not placement references. The @ syntax denotes grouped coordinate constraints; @+name and @-name keep the grouped coordinate on the positive or negative side respectively.

```

pos="x,y"           // start x and y
pos="x,y!"          // pin x and y
pos="x:<coord>"      // start numeric x
pos="x:<coord>!"     // pin x
pos="y:<coord>!"     // pin y
pos="x:<coord>!,y:<coord>" // pin x, start numeric y
pos="x:<coord>,y:<coord>!" // start numeric x, pin y

```

Draw styling is Typst-native. Pass dictionaries or callbacks to draw; edge callbacks receive the merged scope, edge statements, source/sink half-edge records, and the edge index:

```

#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.display-label]
#let source-style(edge) = (stroke: red + 0.5pt)
#let sink-style(edge) = (stroke: blue + 0.5pt)
#draw(layout(g), edge-label: edge-label, source-style: source-style, sink-style: sink-style)

```

Marks can follow graph orientation as a first-class draw option. Put the same mark layer on both halves and set mark-orientation: "edge"; default-oriented edges mark the source half, reversed edges mark the sink half with the marker flipped, and undirected edges suppress the mark.

```

#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: (end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto), scale: 0.75),
)

```

```

    mark-position: "center-if-dangling",
    mark-orientation: "edge",
)
#draw(layout(g), source-style: oriented-arrow, sink-style: oriented-arrow)

```

Physics Edge Styles

`physics.style(..)` returns `source-style`, `sink-style`, and `edge-label` callbacks for `draw`. The default source/sink strokes use darker/lighter halves to encode the graph source/sink split. Fermion map entries marked with `fermion-flow` receive one particle-flow arrow on the main edge using `mark-orientation: "edge"`; paired edges follow `edge.orientation`, dangling half edges place the arrow in the middle of the visible half edge, and undirected edges omit the arrow. This is independent of momentum arrows.

Set `momentum-arrows: true` to draw centered parallel arrow decorations while the main edge remains connected to the nodes. The arrows flow from source to sink independently of the DOT `dir/orientation` value; there is one momentum marker per edge, even though the base edge is internally styled as source and sink halves. On paired edges the marker lives on the sink half; on dangling edges it lives on the existing source or sink half-edge. The decoration defaults to a plain black 0.55pt stroke and a scaled CeTZ "barbed" arrowhead, so it does not inherit the base particle stroke. The arrow length is capped by both `momentum-arrow-length` and `momentum-arrow-ratio`, so the shorter one wins. When the layout provides an `edge-label` position, the arrow offset is signed so the momentum marker is drawn on the same side of the edge as that label. Use `momentum-arrow-offset` to set the normal displacement. `momentum-arrow-mark: auto` uses the default barbed marker, `momentum-arrow-mark: none` draws only the momentum line, and any other value overrides the CeTZ mark style, for example "stealth" or (end: "barbed", scale: 1.6). Only the end mark is kept so source/sink splitting cannot create a second momentum head.

Optional labels can be built from edge metadata with `show-edge-index`, `show-half-edge-index`, `show-particle`, and, for explicit momentum fields, `show-momentum`. `show-half-edge-index` only emits a value for dangling half edges.

```

#import "../src/lib.typ": draw, graph, layout, physics
#import graph: build, dot, edge, edges, node, nodes, parse, sink, source

```

```

#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <fermion-main>,
    sink(<c>),
    statements: (particle: "fermion", id: "7"),
  )
  edge(
    source(<c>),
    <fermion-out>,
    statements: (particle: "fermion", id: "8"),
  )
},
name: "physics",
)
#let callbacks = physics.style(
  momentum-arrows: true,
  show-edge-index: true,
  show-particle: true,
)
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)

```

Set edge-offset on draw, or offset in a source-style/sink-style dictionary, to draw a fitted parallel path. Style callbacks may also return an array of dictionaries; the layers are drawn in order on the same graph edge, so parallel strokes do not require duplicate graph edges.

```
#let base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
)
#let offset-style(edge) = (
  offset: 0.18,
  length: 1.4,
  ratio: 0.5,
  resolve-length: "min",
  stroke: (paint: rgb("#2f6f4e"), thickness: 1pt, cap: "round"),
)
#let source-style(edge) = (base-style(edge), offset-style(edge))
#let sink-style(edge) = (base-style(edge), offset-style(edge) + (mark: (end: ">")))
```

The parallel path is applied to the base edge geometry before patterns and other decorations; node outlets then trim the shifted path, so shifted paths still start and end outside fitted node circles. Add edge-length or length to center-trim the shifted path to a fixed arc length, and add edge-ratio or ratio to cap it by a fraction of the base edge length. edge-resolve-length

resolve-length decides how to combine both limits "min"/"shorter" (default), "max"/"longer",

"length"/"fixed", "ratio"/"relative", "none"/"full", or a function receiving (base-length, length, ratio). Set offset-side: "label" on an offset layer to choose the sign of offset so the layer is drawn on the same side of the curve as the edge label.

source-style-eval, sink-style-eval, and label-eval are ordinary kebab-case edge metadata for downstream renderers or DOT export. They can be set as direct arguments on graph.build or edge, instead of being manually nested under edge-statements or per-edge statements:

```
#let g = build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label-eval: "(text(fill: rgb(\"#{color}\")))[{label}]",
    statements: (color: "0055ff", label: "a-c"),
  )
},
name: "demo",
source-style-eval: "(stroke: red + 0.5pt)",
sink-style-eval: "(stroke: blue + 0.5pt)",
)
```

graph.build also accepts comma-separated items:

```
#let g = build(
  node(<a>),
  node(<b>),
  edge(
    source(<a>, compass: "e"),
    <ab>,
    sink(<b>, compass: "w"),
    label: [ab],
    statements: (color: "0055ff", label: "ab"),
  ),
name: "demo",
statements: (full_num: "x + y"),
node-statements: (shape: "circle"),
edge-statements: (
  color: "000000",
)
```

```

    display-label: "{label}",
  ),
)

```

Graph Queries

`graph.info(g)` returns graph metadata. `nodes(g)` returns node records, and `edges(g)` returns edge records. Pass subgraph: `sg` to filter nodes or edges by an subgraph object.

`graph.join(left, right, key: "statement")` joins matching dangling half edges. The key is read from half-edge data and can be "statement", "compass", or "id".

`graph.cycles(g)` returns subgraph objects for a cycle basis. `graph.forests(g)` returns subgraph objects for spanning forests.

Layout Model

layout starts from a traversal-tree placement, then optimizes the positions of graph nodes and edge control points. The initial tree spacing is

$$L = \lambda \sqrt{\frac{WH}{\max(n, 1)}}$$

with horizontal spacing $\tau_x L$ and vertical spacing $\tau_y L$. Here λ is length-scale, W is viewport-w, H is viewport-h, τ_x is tree-dx, and τ_y is tree-dy. These fields set the geometry scale for both layout modes.

For deterministic, non-iterative placement, use `layout-algo: "tree"` or `layout-algo: "dot"`. "tree" places a traversal forest by levels. "dot" uses a directed layered placement for acyclic inputs, falling back to tree placement when the selected edges contain a cycle. Both modes also work on a subgraph:

```

#let g = parse("digraph partial { a -> b; b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree", subgraph: tree)
#draw(g)

```

Only nodes touched by the selected subgraph are placed. Edge control points are then resolved from the current node positions, so edges outside the selected subgraph are drawn as straight lines unless their own position constraints say otherwise.

`layout-algo: "force"` and `layout-algo: "anneal"` also accept subgraph. For these iterative modes, nodes and edge control points outside the selected subgraph stay fixed and act as boundary points while the selected subgraph is optimized.

Set `layout-nodes: "fixed"` to keep every node at its current pos for this layout pass and move only edge control points. With subgraph, only edges in the selected subgraph are moved; all other edge control points keep their current positions. The fixed-node policy is temporary; the returned graph stores the resulting coordinates as `pos`, but it does not turn them into persistent `pin` constraints. This is useful after one node-placement pass when a later pass should route or relax selected edges without disturbing the node layout:

```

#let g = parse("digraph partial { a [pos=\"0,0\"]; b [pos=\"4,0\"]; c [pos=\"8,0\"]; a -> b; b -> c }").at(0)
#let first = subgraph.bits(g, (true, true, false, false))
#let g = layout(g, layout-algo: "tree", layout-nodes: "fixed", subgraph: first)
#draw(g)

```

The shared spring/charge model uses the following coefficients:

$$c_{vv} = \beta L^2$$

$$c_{ev} = \beta \gamma_{ev} L^2$$

$$c_{ee} = \beta \gamma_{ee} L^2$$

$$c_{center} = \beta g_{center} L^2$$

$$c_{\text{dangling}} = \beta \gamma_{\text{dangling}} L^2$$

The Typst parameter names are beta for β , gamma-ev for γ_{ev} , gamma-ee for γ_{ee} , g-center for g_{center} , and gamma-dangling for γ_{dangling} . The spring stiffness k is k-spring, and the softening constant ε is eps.

In layout-algo: "anneal", linnest minimizes an energy:

$$E = \sum_{i < j} \frac{1}{2} \frac{c_{\text{vv}}}{d(v_i, v_j) + \varepsilon} + \sum_{i, e} \frac{c_{\text{ev}}}{d(v_i, e) + \varepsilon} + \sum_{(v, e) \text{ incident}} \frac{1}{2} k (\ell_e - d(v, e))^2$$

$$+ \sum_{\text{local edge pairs}} \frac{1}{2} \frac{c_{\text{ee}}}{d(e_i, e_j) + \varepsilon} + \sum_{\text{dangling pairs}} \frac{1}{2} \frac{c_{\text{dangling}}}{d(e_i, e_j) + \varepsilon} + \sum_i \frac{1}{2} \frac{c_{\text{center}}}{d(v_i, 0) + \varepsilon} + p_{\text{cross}} N_{\text{cross}}$$

.

Here p_{cross} is crossing-penalty and N_{cross} is the number of detected edge crossings. temp, step, seed, steps, epochs, cool, accept-floor, step-shrink, and incremental-energy belong to this simulated annealing mode. crossing-penalty is also anneal-only; force mode does not currently add a crossing force.

In layout-algo: "force", linnest applies the direct forces corresponding to the same vertex-vertex, edge-vertex, incidence spring, local edge-edge, dangling-edge, and center terms. step is the integration step, delta clamps per-step movement, steps and epochs set the iteration budget, cool shrinks the step after each epoch, and early-tol stops when movement is small. z-spring and z-spring-growth are force-only helpers: the integrator gives points temporary z coordinates to break overlaps and pulls them back toward the 2D plane.

directional-force is applied in both modes as an extra bias derived from pin/port direction constraints.

After either graph layout mode, labels are relaxed separately. If L_l is the label target distance and q_l is the label repulsion strength, then $L_l = \alpha_l L$ and $q_l = \beta_l L^2$. The Typst names are label-length-scale for α_l and label-charge for β_l . label-spring is the spring constant pulling each label toward its target. label-steps, label-step, label-early-tol, and label-max-delta-scale control the label relaxation iteration.

Subgraphs

Subgraph objects are opaque zero-copy values.

- `subgraph.label(g, label)` constructs a subgraph from a base62 label.
- `subgraph.bits(g, bits)` constructs a subgraph from a boolean hedge array.
- `subgraph.compass(g, compass)` selects half edges with a DOT compass point.
- `subgraph.to-label(sg)` returns the base62 label.
- `subgraph.hedges(sg)` returns included hedge indices.
- `subgraph.contains(sg, hedge)` tests hedge membership.

Generated Reference

graph

- `build()`
- `cycles()`
- `dot()`
- `edge()`
- `edge-data()`
- `edges()`
- `eval-fields()`
- `forests()`
- `group()`
- `info()`
- `join()`
- `map()`
- `node()`
- `node-data()`

- nodes()
- parse()
- pos()
- sink()
- source()
- update-edge-data()
- update-node-data()

build

Build one graph object from a stream of node and edge items.

Use node(), source(), sink(), and edge() to create graph items. Positional items may be passed as comma-separated arguments or yielded from a Typst code block.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), <e>,
graph.sink(<b>))
}, name: "demo")
#graph.info(g).name
```

demo

Parameters

```
build(
  ..items: array,
  name: none string,
  payload: any,
  statements: dictionary,
  edge-statements: dictionary,
  source-style-eval: none string,
  sink-style-eval: none string,
  label-eval: none string,
  node-statements: dictionary,
  nodes: array,
  edges: array
) -> bytes
```

..items array

Node and edge items returned by node() and edge().

name none or string

Graph name.

Default: none

payload any

Opaque graph payload. Any CBOR-encodable Typst value.

Default: none

statements dictionary

Flat graph statements. Used by DOT; values cannot nest.

Default: (:)

edge-statements dictionary

Flat default edge statements. Used by DOT; values cannot nest. String values may use {name} placeholders that expand from each edge's merged statement dictionary.

Default: (:)

source-style-eval none or string

Default always-evaluated Typst style for source half-edges. This is merged into edge-statements as source-style-eval.

Default: none

sink-style-eval none or string

Default always-evaluated Typst style for sink half-edges. This is merged into edge-statements as sink-style-eval.

Default: none

label-eval none or string

Default always-evaluated Typst edge label template. This is merged into edge-statements as label-eval.

Default: none

node-statements dictionary

Flat default node statements. Used by DOT; values cannot nest.

Default: (:)

nodes array

Additional node items or raw node specs. -> array

Default: ()

edges array

Additional edge items or raw edge specs.

Default: ()

cycles

Return subgraph objects for the graph's cycle basis.

```
#let g = graph.build({  
  graph.node(<a>  
  graph.node(<b>  
  graph.edge(graph.source(<a>), graph.sink(<b>))  
})
```

```
#graph.cycles(g).len()
```

```
0
```

Parameters

`cycles`(graph) -> array

dot

Serialize a graph object to DOT.

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), graph.sink(<b>))  
}, name: "demo")  
#graph.dot(g).contains("digraph demo")
```

```
true
```

Parameters

`dot`(graph) -> string

edge

Create a graph edge item for build().

Positional arguments may contain one source(), one sink(), and optionally one Typst label used as the edge name. The numeric id chooses the edge order and the visible label is label, stored as payload.label.

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), <e>,  
graph.sink(<b>), label: [$p$])  
})
```

Parameters

```
edge(  
  ..args: any,  
  name: none label,  
  id: none int,  
  orientation: string,  
  label: any,  
  payload: any,  
  pos: none dictionary,  
  shift: none string array dictionary,  
  label-pos: none string array dictionary,  
  label-angle: none int float string,  
  bend: none int float string,  
  statements: dictionary,  
  source-style-eval: none string,  
  sink-style-eval: none string,  
  label-eval: none string  
) -> array
```

..args any

Source/sink half-edges and optional edge name.

name none or label

Typst edge name.

Default: none

id none or int

Numeric edge order/index override.

Default: none

orientation string

Edge orientation: "default", "reversed", or "undirected".

Default: "default"

label any

Visible edge label, stored as `payload.label`.

Default: none

payload any

Opaque edge payload. Any CBOR-encodable Typst value.

Default: none

pos none or dictionary

Edge placement.

Default: none

shift none or string or array or dictionary

Drawing shift stored as a statement.

Default: none

label-pos none or string or array or dictionary

Edge label position stored as a statement.

Default: none

label-angle `none` or `int` or `float` or `string`

Edge label angle stored as a statement.

Default: `none`

bend `none` or `int` or `float` or `string`

Edge bend stored as a statement.

Default: `none`

statements `dictionary`

Additional flat edge statements. Used by DOT; values cannot nest.

Default: `(:)`

source-style-eval `none` or `string`

Always-evaluated Typst style for the source half-edge.

Default: `none`

sink-style-eval `none` or `string`

Always-evaluated Typst style for the sink half-edge.

Default: `none`

label-eval `none` or `string`

Always-evaluated Typst edge label template.

Default: `none`

edge-data

Return one named edge's opaque payload.

name is a Typst label such as `<e>` or the corresponding string name.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), <e>,
graph.sink(<b>), payload: (label: [$p]))
})
#graph.edge-data(g, <e>).label
```

p

Parameters

```
edge-data(
  graph_,
  name
) -> any
```

edges

Return edge records, optionally filtered by an subgraph object.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>, compass: "e"),
  graph.sink(<b>))
})
#let east = subgraph.compass(g, "e")
#graph.edges(g, subgraph: east).len()
```

1

Parameters

```
edges(
  graph,
  subgraph
) -> array
```

eval-fields

Evaluate selected fields into opaque payload entries.

Each selected field is read from the record's merged fields dictionary, evaluated in a scope containing those fields, and written to payload.<field>.

```
#let g = graph.build(
  {
    graph.node(<a>)
    graph.node(<b>)
    graph.edge(graph.source(<a>),
    graph.sink(<b>), statements: (mom: "p"))
  },
  edge-statements: (display-label: "$#mom$"),
)
#let g = graph.eval-fields(g, eval-edge-fields:
("display-label",))
#graph.edges(g).first().payload.at("display-label")
```

p

Parameters

```
eval-fields(
  graph_,
  eval-graph-fields: string array,
  eval-node-fields: string array,
  eval-edge-fields: string array,
  eval-source-fields: string array,
  eval-sink-fields: string array,
  eval-mode: string,
  scope: dictionary
) -> bytes
```

eval-graph-fields string or array

Graph statement fields to evaluate into graph.info(g).payload.

Default: ()

eval-node-fields `string` or `array`

Node statement fields to evaluate into `graph.nodes(g).at(i).payload`.

Default: `()`

eval-edge-fields `string` or `array`

Edge statement fields to evaluate into `graph.edges(g).at(i).payload`.

Default: `()`

eval-source-fields `string` or `array`

Source half-edge fields to evaluate into `edge.source.payload`.

Default: `()`

eval-sink-fields `string` or `array`

Sink half-edge fields to evaluate into `edge.sink.payload`.

Default: `()`

eval-mode `string`

Typst eval mode used for string field values.

Default: `"markup"`

scope `dictionary`

Additional Typst names available while evaluating field values.

Default: `(:)`

forests

Return subgraph objects for the graph's spanning forests.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#graph.forests(g).len()
```

1

Parameters

`forests`(graph) -> `array`

group

Create a grouped placement coordinate.

side: "+" keeps the solved coordinate non-negative and side: "-" keeps it non-positive. Groups are layout constraints and therefore require pin placement, which is the `graph.pos` default.

```
#graph.group("right", side: "+")
```

```
(kind: "group", name: "right", side: "+")
```

Parameters

```
group(  
  name,  
  side  
) -> dictionary
```

info

Return graph metadata.

The result has name, global-statements, edge-statements, and node-statements.

```
#let g = graph.build({ graph.node(<a>) }, name:  
"demo")  
#graph.info(g).name
```

```
demo
```

Parameters

```
info(graph) -> dictionary
```

join

Join two graphs by matching dangling half-edge data on key.

Supported key values are "statement", "compass", and "id".

```
#let left = graph.build({  
  graph.node(<a>)  
  graph.edge(graph.sink(<a>, statement: "j"))  
})  
#let right = graph.build({  
  graph.node(<b>)  
  graph.edge(graph.source(<b>, statement: "j"))  
})  
#graph.edges(graph.join(left, right, key:  
"statement")).len()
```

```
1
```

Parameters

```
join(  
  left,  
  right,  
  key  
) -> bytes
```

map

Map graph metadata to new opaque payloads.

The callbacks receive decoded records plus a `fields` dictionary containing merged statements and direct record fields. A callback returns `none` to leave the record unchanged, or `(payload: value)` to set a new opaque payload.

```
#let g = graph.build({ graph.node(<a>) })
#let g = graph.map(g, node: node => (payload:
(label: [A])))
#graph.nodes(g).first().payload.label
```

A

Parameters

```
map(
  graph_,
  graph: none function,
  node: none function,
  edge: none function,
  source: none function,
  sink: none function
) -> bytes
```

graph none or function

Graph metadata callback.

Default: none

node none or function

Node metadata callback.

Default: none

edge none or function

Edge metadata callback.

Default: none

source none or function

Source half-edge metadata callback.

Default: none

sink none or function

Sink half-edge metadata callback.

Default: none

node

Create a graph node item for build().

A Typst label is the node name used by source(), sink(), and pos(). The optional numeric id chooses the node order/index. payload is opaque Typst-side metadata; label is stored as payload.label.

```
#let g = graph.build({
  graph.node(<a>, id: 0, label: [A])
})
#graph.nodes(g).first().name
```

a

Parameters

```
node(  
  ..args,  
  name: none label ,  
  id: none int ,  
  label: any ,  
  payload: any ,  
  pos: none dictionary ,  
  shift: none string array dictionary ,  
  statements: dictionary  
) -> array
```

name none or label

Typst node name for references.

Default: none

id none or int

Numeric node order/index override.

Default: none

label any

Visible node label, stored as `payload.label`.

Default: none

payload any

Opaque node payload. Any CBOR-encodable Typst value.

Default: none

pos none or dictionary

Node placement.

Default: none

shift none or string or array or dictionary

Drawing shift stored as a statement.

Default: none

statements dictionary

Additional flat node statements. Used by DOT; values cannot nest.

Default: (:)

node-data

Return one named node's opaque payload.

name is a Typst label such as <a> or the corresponding string name.

```
#let g = graph.build({ graph.node(<a>, payload:
  (label: [A])) })
#graph.node-data(g, <a>).label
```

A

Parameters

```
node-data(
  graph_,
  name
) -> any
```

nodes

Return node records, optionally filtered by an subgraph object.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
})
#graph.nodes(g).map(node => node.name).join(",
")
```

a, b

Parameters

```
nodes(
  graph,
  subgraph
) -> array
```

parse

Parse one or more DOT digraphs into graph objects.

```
#let graphs = graph.parse("digraph first { a ->
b }")
#graphs.len()
```

1

Parameters

```
parse(
  input,
  eval-graph-fields: string array,
  eval-node-fields: string array,
  eval-edge-fields: string array,
  eval-source-fields: string array,
  eval-sink-fields: string array,
  eval-mode: string,
  scope: dictionary
) -> array
```

eval-graph-fields string or array

Graph statement fields to evaluate into `graph.info(g).payload`.

Default: ()

eval-node-fields `string` or `array`

Node statement fields to evaluate into `graph.nodes(g).at(i).payload`.

Default: `()`

eval-edge-fields `string` or `array`

Edge statement fields to evaluate into `graph.edges(g).at(i).payload`.

Default: `()`

eval-source-fields `string` or `array`

Source half-edge fields to evaluate into `edge.source.payload`.

Default: `()`

eval-sink-fields `string` or `array`

Sink half-edge fields to evaluate into `edge.sink.payload`.

Default: `()`

eval-mode `string`

Typst eval mode used for string field values.

Default: `"markup"`

scope `dictionary`

Additional Typst names available while evaluating field values.

Default: `(:)`

pos

Create a first-class graph placement.

The default mode: `"pin"` turns numeric and grouped coordinates into layout constraints and also makes the coordinates immediately drawable without a layout pass. Use mode: `"start"` when the coordinate should only seed the layout. `ref` may reference a previously created node index and combines with `dx` and `dy`.

```
#graph.pos(x: 0, y: graph.group("row"), mode: "pin")
```

```
(  
  mode: "pin",  
  x: 0,  
  y: (kind: "group", name: "row", side:  
    none),  
)
```

Parameters

```
pos(  
  x,  
  y,  
  ref,  
  dx,  
  dy,  
  mode  
) -> dictionary
```

sink

Create a sink half-edge endpoint.

node may be a node name like <a> or a numeric node index. name gives the half-edge a Typst name; id is a numeric half-edge order/index override; payload is opaque Typst-side metadata.

```
#graph.sink(<b>, name: <h2>, id: 1, payload:  
(kind: "in"), compass: "w")
```

```
(  
  linnest-kind: "sink",  
  node: <b>,  
  name: <h2>,  
  id: 1,  
  payload: (kind: "in"),  
  statement: none,  
  compass: "w",  
)
```

Parameters

```
sink(  
  node,  
  name,  
  id,  
  payload: any,  
  statement,  
  compass  
) -> dictionary
```

payload any

Opaque sink half-edge payload. Any CBOR-encodable Typst value.

Default: none

source

Create a source half-edge endpoint.

node may be a node name like <a> or a numeric node index. name gives the half-edge a Typst name; id is a numeric half-edge order/index override; payload is opaque Typst-side metadata.

```
#graph.source(<a>, name: <h1>, id: 0, payload:  
(kind: "out"), compass: "e")
```

```
(
  linnest-kind: "source",
  node: <a>,
  name: <h1>,
  id: 0,
  payload: (kind: "out"),
  statement: none,
  compass: "e",
)
```

Parameters

```
source(
  node,
  name,
  id,
  payload: any,
  statement,
  compass
) -> dictionary
```

payload `any`

Opaque source half-edge payload. Any CBOR-encodable Typst value.

Default: `none`

update-edge-data

Update one named edge's opaque payload.

name is a Typst label such as <e> or the corresponding string name. update may be a replacement data value or a function (data, edge) => new-data.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), <e>,
graph.sink(<b>))
})
#let g = graph.update-edge-data(g, <e>, (label:
[$p$]))
#graph.edges(g).first().payload.label
```

p

Parameters

```
update-edge-data(
  graph_,
  name,
  update
) -> bytes
```

update-node-data

Update one named node's opaque payload.

name is a Typst label such as <a> or the corresponding string name. update may be a replacement data value or a function (data, node) => new-data.

```
#let g = graph.build({ graph.node(<a>) })
#let g = graph.update-node-data(g, <a>, (label:
[A]))
#graph.nodes(g).first().payload.label
```

A

Parameters

```
update-node-data(
  graph_,
  name,
  update
) -> bytes
```

- layout()

Variables

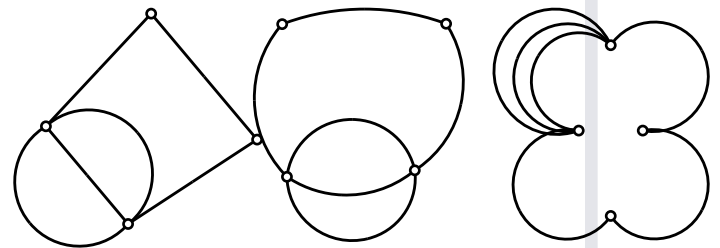
- graph
- curve
- physics
- subgraph

layout

Apply the linest layout pass to a graph object.

This is intentionally a second step: construct or parse a graph first, then call layout. Set layout-algo to "force" for deterministic force integration, "anneal" for simulated annealing, "tree" for a traversal tree placement, or "dot" for a layered directed placement.

```
#let g = graph.parse("digraph partial { a:s ->
b:s;a -> b;a -> b; b:s -> c:s; c:s -> d:s; d:s -
> a:s }").at(0)
#let south = subgraph.compass(g,"s")
#let gf = layout(layout(g, layout-algo:
"force"), layout-algo: "tree", layout-nodes:
"fixed",subgraph:south)
#let ga = layout(g, layout-algo: "anneal")
#let gt = layout(layout(g, layout-algo:
"tree",layout-roots: (2)), layout-algo:
"anneal", layout-nodes: "fixed",gamma-
ee:0.1,gamma-ev:1.5,beta:20,length-scale:0.4)
#grid(columns: 3, gutter: 2cm, draw(gf),
draw(ga), draw(gt))
```



```
#let g = graph.parse("digraph partial { a -> b;
b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree",
subgraph: tree)
#graph.edges(g).map(edge => edge.pos)
```

```
(
  (x: 0.0, y: 1.5),
  (x: 0.0, y: 4.5),
  (x: 0.0, y: 7.5),
  (x: 0.0, y: 4.5),
)
```


Parameters

```
layout(  
  graph: bytes ,  
  subgraph: none bytes ,  
  viewport-w: float ,  
  viewport-h: float ,  
  tree-dx: float ,  
  tree-dy: float ,  
  steps: int ,  
  seed: int ,  
  step: float ,  
  step-shrink: float ,  
  cool: float ,  
  accept-floor: float ,  
  early-tol: float ,  
  temp: float ,  
  delta: float ,  
  beta: float ,  
  k-spring: float ,  
  g-center: float ,  
  epochs: int ,  
  crossing-penalty: float ,  
  gamma-dangling: float ,  
  gamma-ee: float ,  
  directional-force: float ,  
  label-length-scale: float ,  
  label-spring: float ,  
  label-charge: float ,  
  label-steps: int ,  
  label-step: float ,  
  label-early-tol: float ,  
  label-max-delta-scale: float ,  
  gamma-ev: float ,  
  eps: float ,  
  incremental-energy: bool ,  
  layout-algo: string ,  
  layout-nodes: string ,  
  layout-roots: array ,  
  z-spring: float ,  
  z-spring-growth: float ,  
  length-scale: float  
) -> bytes
```

graph bytes

Graph object returned by `graph.build` or `graph.parse`.

subgraph none or bytes

Optional subgraph object to lay out. With "tree" and "dot", other edges are drawn from the resulting node positions as straight lines. With "force" and "anneal", nodes and edges outside the subgraph are fixed boundary points during optimization.

Default: none

viewport-w float

Width of the layout viewport used to derive the natural spring length. Applies to both "force" and "anneal".

Default: 10.0

viewport-h float

Height of the layout viewport used to derive the natural spring length. Applies to both "force" and "anneal".

Default: 10.0

tree-dx float

Horizontal spacing multiplier for the initial traversal-tree placement. Applies before both "force" and "anneal".

Default: 0.9

tree-dy float

Vertical spacing multiplier for the initial traversal-tree placement. Applies before both "force" and "anneal".

Default: 1.2

steps int

Iterations per epoch. In "force" mode this is the number of force integration steps; in "anneal" mode this is the number of proposals per temperature epoch.

Default: `int(sys.inputs.at("steps", default: "30"))`

seed int

Seed for deterministic initialization, force-mode jitter, and annealing proposals. Applies to both modes.

Default: `int(sys.inputs.at("seed", default: "2"))`

step float

Initial movement scale. "force" multiplies computed forces by this value; "anneal" uses it as the proposal step size.

Default: 0.81

step-shrink float

Anneal-only step shrink factor, applied when an epoch's acceptance ratio falls below `accept-floor`.

Default: 0.21

cool float

Cooling factor applied once per epoch. "anneal" cools temp; "force" shrinks step.

Default: 0.85

accept-floor float

Anneal-only acceptance-ratio threshold below which step is shrunk by step-shrink.

Default: 0.15

early-tol float

Force-only early stop threshold for maximum movement in one step. The annealing schedule stores this value but does not currently use it for stopping.

Default: 1e-6

temp float

Anneal-only initial temperature used in the Metropolis acceptance test.

Default: 0.3

delta float

Force-mode maximum movement clamp per point and per step.

Default: 0.4

beta float

Base repulsion strength for vertex-vertex interactions. Also scales gamma-ev, gamma-ee, gamma-dangling, and g-center. Applies to both modes through the shared spring energy.

Default: 50.0

k-spring float

Spring stiffness for node-to-edge incidence lengths. Applies to both modes.

Default: 11.0

g-center float

Centering strength relative to beta. Applies to both modes.

Default: 0.005

epochs int

Number of epochs. Both modes run up to steps iterations inside each epoch.

Default: 30

crossing-penalty float

Anneal-only fixed energy penalty per detected edge crossing. The direct force integrator does not currently add a crossing force.

Default: 30.0

gamma-dangling float

Repulsion for dangling half edges, relative to beta. Applies to both modes through the shared spring energy.

Default: 5.0

gamma-ee float

Local edge-edge repulsion, relative to beta. Applies to both modes.

Default: 0.1

directional-force float

Bias that pushes points in directions implied by pin/port constraints. Applies to both modes.

Default: 5.0

label-length-scale float

Edge-label target offset as a multiple of the graph spring length. Label layout runs after both graph layout modes.

Default: 0.6

label-spring float

Spring strength pulling each label toward its target offset. Label layout runs after both modes.

Default: 23.0

label-charge float

Repulsion strength between labels and graph points, scaled by spring length squared. Label layout runs after both modes.

Default: 3.0

label-steps int

Maximum number of post-layout label relaxation steps. Set to 0 to skip label placement. Applies after both modes.

Default: 20

label-step float

Label relaxation step size. Applies after both modes.

Default: 0.15

label-early-tol float

Label relaxation early stop threshold. Applies after both modes.

Default: 1e-3

label-max-delta-scale float

Label movement clamp as a multiple of spring length. Applies after both modes.

Default: 0.5

gamma-ev float

Edge-vertex repulsion, relative to beta. Applies to both modes.

Default: 0.01

eps float

Softening epsilon used in inverse-square force/energy terms. Applies to both modes.

Default: 1e-4

incremental-energy bool

Whether annealing updates cached energy by local deltas. This is anneal-only; force mode computes direct forces instead of energies.

Default: true

layout-algo string

Layout algorithm. Use "force" for direct force integration, "anneal" for simulated annealing against the spring energy, "tree" for a traversal-tree placement, or "dot" for a layered directed placement.

Default: "force"

layout-nodes string

Node movement policy. "layout" lets the layout algorithm move nodes. "fixed" keeps every node at its current position for this layout pass and only moves edge control points. With subgraph, only edges in the subgraph are moved; other edge control points stay at their current positions.

Default: "layout"

layout-roots array

Ordered node indices used as preferred roots for "tree" and "dot". Roots outside the selected node set are ignored. Remaining components are laid out afterward in graph order.

Default: ()

z-spring float

Force-only spring pulling temporary z coordinates back toward the layout plane. Use with z-spring-growth to help separate overlapping points during integration.

Default: 0.05

z-spring-growth float

Force-only per-epoch multiplier for z-spring.

Default: 1.3

length-scale float

Natural spring-length multiplier. This scales the graph's preferred edge length and the repulsive coefficients derived from it. Applies to both modes.

Default: 0.5

graph module

Graph namespace.

Graph objects are opaque zero-copy values. Keep them opaque and pass them back to this module, subgraph, or layout().

curve module

Curve namespace.

Helpers for splitting and drawing Bezier edge geometry.

physics module

Physics edge-style namespace.

Reusable particle-line styles and callbacks for diagrams drawn with draw.

subgraph module

Subgraph namespace.

Subgraph objects are opaque zero-copy values. They can be passed to graph.nodes and graph.edges.

physics

- `dangling-half-edge-index()`
- `edge-entry()`
- `edge-index()`
- `edge-label()`
- `eval-template()`
- `interpolate-template()`
- `label-content()`
- `momentum-value()`
- `particle-name()`
- `sink-stroke()`
- `sink-style()`
- `source-stroke()`
- `source-style()`
- `stroke-style()`
- `style()`
- `style-dict()`
- `text-value()`

Variables

- `massive`
- `massless`
- `dashed`
- `dotted`
- `fermion-arrow-mark`
- `fermion-flow`
- `wave`
- `coil`
- `zigzag`
- `default-edge`
- `default-map`
- `momentum-arrow-defaults`

dangling-half-edge-index

Return the exposed half-edge index for dangling edges only.

Parameters

`dangling-half-edge-index`(edge) -> `none` `any`

edge-entry

Return the style-map entry for an edge.

Parameters

```
edge-entry(  
  edge,  
  map,  
  default  
) -> dictionary
```

edge-index

Return the edge index used by optional edge labels. The DOT `id` statement wins over the renderer-local `eid`.

Parameters

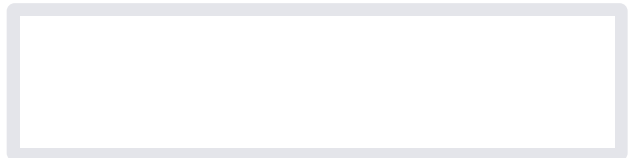
```
edge-index(  
  edge,  
  fields  
) -> none any
```

edge-label

Edge-label callback.

By default this preserves explicit label-eval, display-label, label-template, label, and particle-map labels. Set any show-* option to build a label from selected metadata instead:

```
#let callbacks = physics.style(  
  momentum-arrows: true,  
  show-edge-index: true,  
  show-particle: true,  
)
```



Parameters

```
edge-label(  
  edge,  
  map,  
  default,  
  typst-fields,  
  scope,  
  show-momentum,  
  show-edge-index,  
  show-half-edge-index,  
  show-particle,  
  momentum-fields,  
  edge-index-fields,  
  momentum-prefix,  
  edge-index-prefix,  
  half-edge-index-prefix,  
  particle-prefix,  
  label-separator,  
  ],  
  label-size,  
  label-fill  
) -> none content
```

eval-template

Evaluate a string-valued style template after placeholder interpolation.

Parameters

```
eval-template(  
  template,  
  edge,  
  map,  
  scope  
) -> any
```


interpolate-template

Replace {field} placeholders in a string-valued style template. {{ and }} produce literal braces.

Parameters

```
interpolate-template(  
    template,  
    edge  
) -> none string
```

label-content

Convert a style label value to content. In mode: "eval", string values are evaluated after placeholder interpolation.

Parameters

```
label-content(  
    value,  
    edge,  
    mode,  
    map,  
    scope  
) -> none content
```

momentum-value

Return the momentum field used by optional edge labels.

Parameters

```
momentum-value(  
    edge,  
    fields  
) -> none any
```

particle-name

Return an edge's particle name, stripping the quotes often present in DOT statement values.

Parameters

```
particle-name(edge) -> none string
```

sink-stroke

Sink-half stroke helper. The default lightening makes the two halves encode the graph's source/sink split without requiring arrowheads.

Parameters

```
sink-stroke(  
    c,  
    thickness,  
    dash,  
    lighten  
) -> dictionary
```

sink-style

Style callback for the sink half edge.

`momentum-arrows: true` adds one centered black CeTZ-mark decoration toward the sink node, so momentum arrows always flow from source to sink independently of `edge.orientation`.

Parameters

```
sink-style(  
  edge,  
  map,  
  default,  
  typst-fields,  
  scope,  
  orientation-split,  
  momentum-arrows,  
  momentum-arrow-offset,  
  momentum-arrow-length,  
  momentum-arrow-ratio,  
  momentum-arrow-stroke,  
  momentum-arrow-mark  
) -> dictionary array
```

source-stroke

Source-half stroke helper.

Parameters

```
source-stroke(  
  c,  
  thickness,  
  dash  
) -> dictionary
```

source-style

Style callback for the source half edge.

`momentum-arrows: true` adds one centered black CeTZ-mark decoration while the main edge remains drawn with its normal particle style.

Parameters

```
source-style(  
  edge,  
  map,  
  default,  
  typst-fields,  
  scope,  
  orientation-split,  
  momentum-arrows,  
  momentum-arrow-offset,  
  momentum-arrow-length,  
  momentum-arrow-ratio,  
  momentum-arrow-stroke,  
  momentum-arrow-mark  
) -> dictionary array
```

stroke-style

Build a rounded CeTZ stroke dictionary accepted by `linnest.draw`.

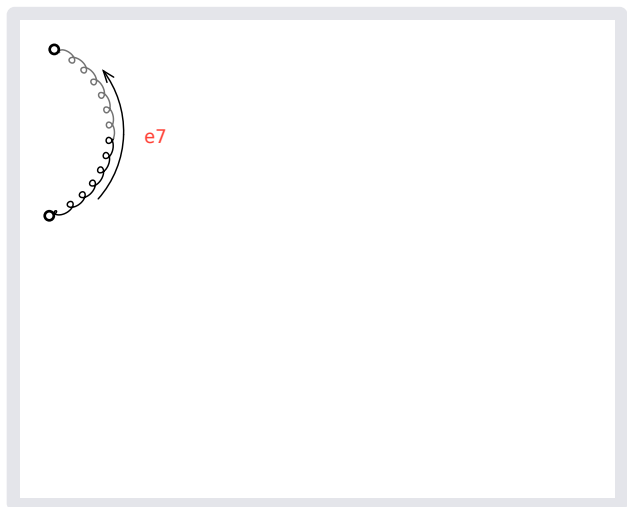
Parameters

```
stroke-style(  
  c,  
  thickness,  
  dash  
) -> dictionary
```

style

Bundle source-style, sink-style, and edge-label callbacks with shared options for `linnest.draw`.

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), <g-edge>,  
graph.sink(<b>), statements: (particle: "g", id:  
"7"))  
},  
  name: "physics",  
)  
#let callbacks = physics.style(momentum-arrows:  
true, show-edge-index: true)  
#draw(  
  layout(g),  
  source-style: callbacks.source-style,  
  sink-style: callbacks.sink-style,  
  edge-label: callbacks.edge-label,  
)
```



Parameters

```
style(  
  map,  
  default,  
  typst-fields,  
  scope,  
  orientation-split,  
  momentum-arrows,  
  momentum-arrow-offset,  
  momentum-arrow-length,  
  momentum-arrow-ratio,  
  momentum-arrow-stroke,  
  momentum-arrow-mark,  
  show-momentum,  
  show-edge-index,  
  show-half-edge-index,  
  show-particle,  
  momentum-fields,  
  edge-index-fields,  
  momentum-prefix,  
  edge-index-prefix,  
  half-edge-index-prefix,  
  particle-prefix,  
  label-separator,  
  ],  
  label-size,  
  label-fill  
) -> dictionary
```

style-dict

Convert a style value to a dictionary. In mode: "eval", string values are evaluated after placeholder interpolation.

Parameters

```
style-dict(  
  value,  
  edge,  
  mode,  
  map,  
  scope  
) -> dictionary
```

text-value

Convert DOT-ish values to plain text.

Parameters

```
text-value(value) -> string
```

massive length

Conventional massive-particle stroke thickness.

massless length

Conventional massless-particle stroke thickness.

dashed array

Dash pattern used for scalar-style lines.

dotted string

Dotted dash style used for ghost-style lines.

fermion-arrow-mark dictionary

CeTZ marker used for fermion particle-flow arrows on the main edge.

fermion-flow dictionary

Mark an edge-map entry as a fermion so the main edge receives one particle-flow arrow that follows edge.orientation. This is separate from optional momentum arrows.

wave dictionary

Photon-style wave pattern.

coil dictionary

Gluon-style coil pattern.

zigzag dictionary

Weak-boson-style zigzag pattern.

default-edge dictionary

Fallback style entry used when an edge has no known particle type.

default-map dictionary

Small built-in particle map for standalone physics diagrams. GammaLoop generated edge-style.typ files pass their model-specific map to style.

momentum-arrow-defaults dictionary

Default style for source-to-sink momentum arrow layers.

draw

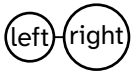
- draw()
- edge-halves()
- to-cetz-edge-halves()

draw

Draw a graph object with CeTZ.

The graph must already have positions, either from layout or from explicit pos values passed to graph node or edge items. Paired edges without an explicit edge position use the midpoint of their endpoint nodes. Node and edge Typst style dictionaries or callbacks are evaluated and forwarded to CeTZ.

```
#let positioned = graph.build({
  graph.node(<left>, label: [left], pos: graph.pos(x: 0, y: 0))
  graph.node(<right>, label: [right], pos: graph.pos(ref: <left>, dx: 2.5, dy: 0))
  graph.edge(graph.source(<left>), graph.sink(<right>))
  graph.edge(graph.source(<right>), <right-out>, pos: graph.pos(ref: <right>, dx: 0.9, dy: 0.7))
},
  name: "positioned",
)
#draw(positioned, source-style: (stroke: black + 0.7pt), sink-style: (stroke: black + 0.7pt))
```

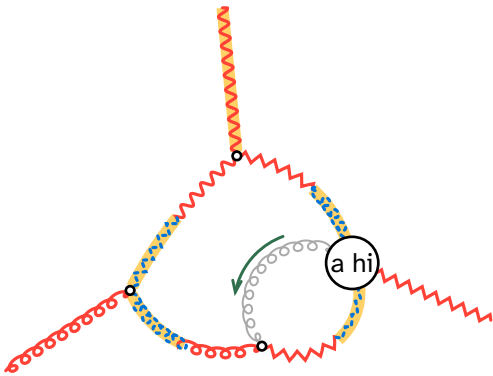


```
#let parallel-base-edge = 0
#let source-patterns = (
  "coil",
  "zigzag",
  "coil",
  "wave",
  "zigzag",
  "coil",
  "wave",
  "zigzag",
)
#let sink-patterns = (
  "coil",
  "zigzag",
  "coil",
  "zigzag",
  "coil",
  "wave",
  "coil",
  "wave",
)
#let parallel-layer(edge, mark: none) = if edge.eid == parallel-base-edge {
  (
    offset: -0.5,
    length: 1.5,
    ratio: 0.5,
    resolve-length: "min",
    stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
    mark: mark,
  )
} else { none }
#let stack(base, layer) = if layer == none { base } else { (base, layer) }
#let source-stroke(edge) = if edge.eid == parallel-base-edge {
  (paint: gray, thickness: 0.75pt, cap: "round")
} else {
  (paint: red, thickness: 1.2pt, cap: "round")
}
#let sink-stroke(edge) = if edge.eid == parallel-base-edge {
  (paint: gray, thickness: 0.75pt, cap: "round")
} else {
  (paint: blue, thickness: 1.2pt, cap: "round", dash: "dotted")
}
#let source-style(edge) = {
  let base = (
    stroke: source-stroke(edge),
    pattern: source-patterns.at(edge.eid),
    pattern-amplitude: 0.18,
    pattern-wavelength: 0.55,
    pattern-coil-longitudinal-scale: 1.6,
  )
  stack(base, parallel-layer(edge))
}
#let sink-style(edge) = {
  let base = (
    stroke: sink-stroke(edge),
    pattern: sink-patterns.at(edge.eid),
    pattern-amplitude: 0.18,
    pattern-wavelength: 0.55,
    pattern-coil-longitudinal-scale: 1.6,
  )
  stack(base, parallel-layer(edge, mark: (end: (symbol: "straight"), scale: 0.75)))
}
```

```

#let g = graph.build({
  graph.node(<a>, label: [a hi])
  graph.node(<c>)
  graph.node(<d>)
  graph.node(<e>)
  graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), pos: graph.pos(x: 0, y: 0.75, mode: "pin"))
  graph.edge(graph.source(<c>), <ca>, graph.sink(<a>), compass: "e")
  graph.edge(graph.source(<c>), <cd>, graph.sink(<d>), compass: "e")
  graph.edge(graph.source(<e>), <ed>, graph.sink(<d>), compass: "e")
  graph.edge(graph.source(<e>), <ea>, graph.sink(<a>), compass: "e")
  graph.edge(graph.source(<d>), <d-out>)
  graph.edge(graph.source(<e>, compass: "e"), <e-out>)
  graph.edge(graph.source(<a>), <a-out>)
},
  name: "demo",
)
#let layed-out = layout(g)
#let east = subgraph.compass(layed-out, "e")
#draw(layed-out, subgraph: east, source-style: source-style, sink-style: sink-style)

```



```

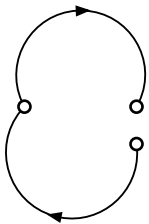
#let p = graph.build({
  graph.node(<pa>, label: [a], pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc1>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc2>, label: [c], pos: graph.pos(y: 0, mode: "pin"))
  graph.edge(graph.source(<pa>), <e0>, graph.sink(<pc>))
  graph.edge(graph.source(<pc2>), <e1>, graph.sink(<pc1>))
},
  name: "parallel demo",
)
#let parallel-edge-style(edge) = (
  offset: -0.5,
  length: 1.4,
  ratio: 0.5,
  resolve-length: "min",
  stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
)
#let focused-base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
  pattern: "coil",
  pattern-amplitude: 0.14,
  pattern-wavelength: 0.45,
  pattern-coil-longitudinal-scale: 1.5,
)
#let focused-source-style(edge) = (focused-base-style(edge), parallel-edge-style(edge))
#let focused-sink-style(edge) = (focused-base-style(edge), parallel-edge-style(edge) + (
  mark: (end: ">"),
))
#draw(layout(p, g-center: 0.004, label-steps: 0,), unit: 1.25, source-style: focused-source-style, sink-

```

```
style: focused-sink-style)
```



```
#let g = graph.build({
  graph.node(<a>, pos: graph.pos(x: 0, y: 0, mode: "pin"))
  graph.node(<c>, pos: graph.pos(x: 3, y: 0, mode: "pin"))
  graph.node(<d>, pos: graph.pos(x: 3, y: -1, mode: "pin"))
  graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), orientation: "default")
  graph.edge(graph.source(<a>), <ad>, graph.sink(<d>), orientation: "reversed")
},
  name: "oriented marks",
)
#let arrow = (
  end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto),
  scale: 0.75,
)
#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: arrow,
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
#draw(layout(g), unit: 1.4, source-style: oriented-arrow, sink-style: oriented-arrow)
```



Parameters

```
draw(  
  graph: bytes ,  
  scope: dictionary ,  
  unit: int float length ratio ,  
  title: none auto content string ,  
  subgraph: none bytes ,  
  debug: bool int ,  
  node-radius: auto int float array ,  
  node-min-radius: int float ,  
  node-label-padding: int float ,  
  node-fill: any ,  
  node-stroke: any ,  
  node-outset: auto int float ,  
  node-label-style: dictionary ,  
  node-style: dictionary function none ,  
  node-label: auto content string function none ,  
  edge-stroke: any ,  
  edge-offset: int float ,  
  edge-length: none int float ,  
  edge-ratio: none int float ,  
  edge-resolve-length: string function ,  
  edge-accuracy: float ,  
  edge-optimize: bool ,  
  source-style: dictionary array function none ,  
  sink-style: dictionary array function none ,  
  edge-label: content string function none ,  
  edge-omega: float ,  
  edge-trim-accuracy: float ,  
  padding: none int float array dictionary ,  
  debug-edge-radius: int float ,  
  debug-edge-fill: any ,  
  debug-edge-stroke: any ,  
  debug-edge-label-fill: any ,  
  subgraph-edge-style: dictionary ,  
  subgraph-edge-underlay: bool  
) -> content
```

graph bytes

Graph object with positions from layout or explicit graph API pos fields.

scope dictionary

Additional values merged into node and edge callback dictionaries.

Default: (:)

unit int or float or length or ratio

Coordinate length for one graph-layout unit. Numbers are interpreted as em.

Default: 1

title `none` or `auto` or `content` or `string`

Optional title displayed above the diagram. Use auto for the graph name.

Default: `none`

subgraph `none` or `bytes`

Optional subgraph whose half-edges are shaded.

Default: `none`

debug `bool` or `int`

Debug level. 1 enables CeTZ canvas debug; 2 also marks edge positions.

Default: `false`

node-radius `auto` or `int` or `float` or `array`

Default CeTZ node radius. Use auto to fit the node label.

Default: `auto`

node-min-radius `int` or `float`

Minimum radius used when node-radius is auto.

Default: `0.16`

node-label-padding `int` or `float`

Extra canvas-unit padding around labels when node-radius is auto.

Default: `0.08`

node-fill `any`

Default CeTZ node fill.

Default: white

node-stroke `any`

Default CeTZ node stroke.

Default: black

node-outset `auto` or `int` or `float`

Edge clearance from node centers. auto uses each node circle radius. Increase this when labels extend beyond the circle.

Default: `auto`

node-label-style dictionary

Default node-label style forwarded to `cetz.draw.content`.

Default: `(:)`

node-style dictionary or function or none

Node style dictionary or callback. A callback receives node data.

Default: `(:)`

node-label auto or content or string or function or none

Node label content or callback. auto uses the node name.

Default: `auto`

edge-stroke any

Default CeTZ edge stroke.

Default: `0.1em`

edge-offset int or float

Default normal offset for edge paths. Applied to the base edge geometry before patterns; node outsets then trim the shifted path.

Default: `_edge-geometry-defaults.offset`

edge-length none or int or float

Maximum visible arc length for centered parallel edge paths. none keeps the full shifted path.

Default: `_edge-geometry-defaults.length`

edge-ratio none or int or float

Maximum visible fraction of the base edge length for centered parallel edge paths. Combined with `edge-length` according to `edge-resolve-length`.

Default: `_edge-geometry-defaults.ratio`

edge-resolve-length string or function

Resolve `edge-length` and `edge-ratio`. Accepted string values are "min"/"shorter", "max"/"longer", "length"/"fixed", "ratio"/"relative", or "none"/"full". A function receives (`base-length`, `length`, `ratio`).

Default: `_edge-geometry-defaults.resolve-length`

edge-accuracy float

Arc-length accuracy for fitted parallel edge paths.

Default: `_edge-geometry-defaults.accuracy`

edge-optimize bool

Let Kurbo optimize the fitted parallel path.

Default: `_edge-geometry-defaults.optimize`

source-style dictionary or array or function or none

Source half-edge style dictionary, array of layer dictionaries, or callback. `mark-position`: "center-if-dangling" keeps an end marker at the paired-edge split point while centering it on dangling half edges. `mark-orientation`: "edge" makes a mark follow edge.orientation instead of raw path direction; reversed edges move the mark to the sink half and flip it, while undirected edges suppress it.

Default: `(:)`

sink-style dictionary or array or function or none

Sink half-edge style dictionary, array of layer dictionaries, or callback. A callback receives edge data. `mark-position`: "center-if-dangling" has the same dangling-edge behavior as for `source-style`, and `mark-orientation`: "edge" participates in the same orientation-aware mark placement.

Default: `(:)`

edge-label content or string or function or none

Edge label content or callback. A callback receives edge data.

Default: `none`

edge-omega float

Hobby curl used at the endpoints of paired edge curves.

Default: `1.0`

edge-trim-accuracy float

Arc-length accuracy for trimming edge curves at node outsets.

Default: `0.001`

padding none or int or float or array or dictionary

CeTZ canvas padding.

Default: `0.4`

debug-edge-radius `int` or `float`

Radius for edge-position markers shown at `debug >= 2`.

Default: `0.08`

debug-edge-fill `any`

Fill for edge-position markers shown at `debug >= 2`.

Default: `rgb("#ff9f1c")`

debug-edge-stroke `any`

Stroke for edge-position markers shown at `debug >= 2`.

Default: `rgb("#d72638") + 0.35pt`

debug-edge-label-fill `any`

Label fill for edge-position markers shown at `debug >= 2`.

Default: `rgb("#7a1020")`

subgraph-edge-style `dictionary`

CeTZ style used to shade half-edges included in subgraph.

Default: `(stroke: rgb("#ffd166") + 4.5pt)`

subgraph-edge-underlay `bool`

Draw subgraph shading below the normal half-edge style.

Default: `true`

edge-halves

Split a laid-out graph edge into source and sink half-edge paths.

The returned dictionary has `source`, `sink`, and `curve`. The split point is the edge layout point, so the two half-edges join smoothly there.

Parameters

```
edge-halves(  
  edge,  
  nodes,  
  omega,  
  source-outset,  
  sink-outset,  
  accuracy  
) -> dictionary
```

to-cetz-edge-halves

Draw the two halves of a laid-out graph edge through CeTZ.

source-style applies from the source node to the edge layout point, and sink-style applies from the edge layout point to the sink node.

Parameters

```
to-cetz-edge-halves(  
  edge,  
  nodes,  
  unit,  
  omega,  
  source-outset,  
  sink-outset,  
  accuracy,  
  source-style,  
  sink-style  
) -> content
```

subgraph

- bits()
- compass()
- contains()
- hedges()
- label()
- to-label()

bits

Construct an subgraph object from a boolean hedge array.

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), graph.sink(<b>))  
})  
#let first = subgraph.bits(g, (true, false))  
#subgraph.contains(first, 0)
```

true

Parameters

```
bits(  
  graph,  
  bits  
) -> bytes
```

compass

Construct an subgraph object from a DOT compass point such as "n" or "s".

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), compass: "e"),  
  graph.sink(<b>))  
})  
#subgraph.hedges(subgraph.compass(g, "e")).len()
```

1

Parameters

```
compass(  
  graph,  
  compass  
) -> bytes
```

contains

Test whether an subgraph object includes a hedge.

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), graph.sink(<b>))  
})  
#subgraph.contains(subgraph.bits(g, (true,  
false)), 0)
```

true

Parameters

```
contains(  
  subgraph,  
  hedge  
) -> bool
```

hedges

Return the hedge indices included in an subgraph object.

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), graph.sink(<b>))  
})  
#subgraph.hedges(subgraph.bits(g, (true,  
false)))
```

(0,)

Parameters

```
hedges(subgraph) -> array
```

label

Construct an subgraph object from a base62 label.

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), compass: "e"),  
  graph.sink(<b>))  
})  
#let east = subgraph.compass(g, "e")  
#let same = subgraph.label(g, subgraph.to-  
label(east))  
#subgraph.hedges(same).len()
```

1

Parameters

```
label(  
  graph,  
  label  
) -> bytes
```

to-label

Convert an subgraph object to its base62 label.

```
#let g = graph.build({  
  graph.node(<a>)  
  graph.node(<b>)  
  graph.edge(graph.source(<a>), graph.sink(<b>))  
})  
#subgraph.to-label(subgraph.bits(g, (true,  
false)))
```

1

Parameters

```
to-label(subgraph) -> string
```